Open Project QCG, IITR

Quantum Random Walks (in Bond Pricing)

Assignment 1

Question 1

Write a Python function **price_zero_coupon_bond** (**face_value, yield_rate, time_to_maturity**) that calculates the present value (current price) of a zero-coupon bond.

- Use continuous compounding for the discounting calculation.
- Call your function to price a 1-year Treasury bill with a $1,000 face value at a 5% yield.
- Call your function again to price a 5-year Treasury note with a $1,000 face value at a 5% yield.

Question 2

Prove this inverse relationship computationally. Using your function from Question 1, define a 10-year zero-coupon bond with a $100 face value.

- Generate a sequence of interest rates (yields) from 1% to 20%, in increments of 1%.
- Iterate through these yields and calculate the bond's price at each rate.
- Print a formatted table (using a Pandas DataFrame or formatted standard output) with two columns: Yield (%) and Bond Price ($).

Question 3

Extend your script from Question 2 to use matplotlib (or another plotting library). Instead of looking at just one bond, you will visualize three on the same graph.

- Calculate the prices across the 1%–20% yield sweep for **three** different zero-coupon bonds: a 2-year bond, a 10-year bond, and a 30-year bond (all $100 face value).
- Plot all three price-yield curves on a single chart. Put "Yield" on the X-axis and "Bond Price" on the Y-axis. Ensure the plot has a title, labeled axes, and a legend distinguishing the 2Y, 10Y, and 30Y curves.